SentinelOps AI – Autonomous Incident Response & Post-Mortem Copilot

HiDevs × Mastra Hackathon 2026

Track: Incident Response & Post-Mortem Agent

1. Executive Summary

SentinelOps AI is a production-grade, event-driven, AI-powered Incident Response and Post-Mortem Copilot designed to help Site Reliability Engineering (SRE) and DevOps teams detect, diagnose, resolve, and learn from system incidents autonomously.

Modern software systems generate enormous volumes of logs, metrics, and telemetry data. During incidents, engineering teams often spend significant time identifying root causes, searching historical knowledge, coordinating remediation, and manually documenting post-mortems. This increases Mean Time to Resolution (MTTR), causes knowledge loss, and leads to repeated failures.

SentinelOps AI addresses these challenges through an intelligent multi-agent architecture that combines real-time telemetry analysis, institutional memory, safety evaluation, and continuous learning. The platform leverages Mastra for agent orchestration, Qdrant for semantic memory and knowledge retrieval, and Enkrypt AI for safety validation and trustworthy recommendations.

The system continuously learns from every incident and becomes progressively smarter over time, enabling organizations to reduce downtime, preserve operational knowledge, and improve service reliability.

2. Problem Statement

Modern cloud-native applications operate across distributed environments consisting of microservices, containers, APIs, and cloud infrastructure. These systems generate massive volumes of telemetry data every second.

When incidents occur, engineering teams face several challenges:

- Huge amounts of logs and metrics make root cause identification difficult.
- Incident response is often reactive and time-consuming.
- Knowledge from previous incidents is scattered across documents and dashboards.
- Engineers repeatedly solve similar incidents because organizational learning is poor.
- Remediation recommendations can be incomplete or unsafe.
- Post-mortem documentation is largely manual and inconsistent.

These issues increase downtime, reduce productivity, and negatively impact business operations.

There is a need for an intelligent engineering copilot that can detect incidents, retrieve historical knowledge, provide safe recommendations, automate documentation, and continuously learn from every incident.

3. Proposed Solution

SentinelOps AI is an autonomous engineering copilot that continuously monitors logs, metrics, and distributed traces to identify incidents and assist engineers throughout the entire incident lifecycle.

The platform performs:

- Telemetry ingestion and preprocessing
- Anomaly detection and incident severity classification
- Retrieval of similar historical incidents
- Root cause analysis
- Safe remediation recommendations
- Automated post-mortem generation
- Continuous organizational learning

Rather than acting as a simple chatbot, SentinelOps AI functions as a production-grade incident intelligence platform capable of reasoning, remembering, evaluating, and continuously improving.

4. Objectives

The primary objectives of SentinelOps AI are:

1. Reduce Mean Time to Resolution (MTTR)
2. Improve incident diagnosis through historical knowledge retrieval
3. Preserve organizational knowledge and incident history
4. Provide trustworthy and safe remediation recommendations
5. Automate post-mortem documentation
6. Enable continuous learning and self-improvement
7. Improve reliability of cloud-native systems
8. Assist engineering teams through intelligent automation

5. Target Users

SentinelOps AI is designed for:

- Site Reliability Engineers (SREs)
- DevOps Engineers
- Platform Engineers
- Cloud Operations Teams
- Infrastructure Teams
- Engineering Managers
- Production Support Teams

6. Key Features

6.1 Telemetry Ingestion

The platform continuously collects logs, metrics, and traces from:

- Kubernetes Clusters
- Applications and APIs
- Cloud Infrastructure
- Prometheus
- Grafana
- Datadog
- OpenTelemetry
- Jaeger
- AWS CloudWatch

6.2 Anomaly Detection

The system automatically detects:

- Error spikes
- Latency increases
- CPU and memory abnormalities
- Service failures
- Infrastructure anomalies
- Unusual traffic patterns

The system also assigns incident severity levels.

6.3 Historical Incident Retrieval

SentinelOps AI performs semantic search on historical incidents and retrieves:

- Similar incidents
- Previous root causes
- Successful remediation steps
- Runbooks
- Post-mortem reports
- Lessons learned

6.4 Root Cause Analysis

The platform correlates:

- Logs
- Metrics
- Traces
- Service dependencies
- Historical incidents

It then generates probable causes with confidence scores.

6.5 Safe Remediation Recommendations

The platform proposes:

- Recovery actions
- Troubleshooting steps
- Infrastructure runbooks
- Configuration recommendations
- Preventive measures

All recommendations are validated for safety before reaching engineers.

6.6 Automated Post-Mortem Generation

The system automatically generates:

- Incident timeline
- Impact assessment
- Root cause analysis
- Actions taken
- Resolution steps
- Preventive measures
- Lessons learned

6.7 Continuous Learning

Every resolved incident is stored as institutional knowledge so that future incidents can benefit from previous experience.

7. System Architecture

SentinelOps AI follows a modular, event-driven, cloud-native architecture consisting of:

1. **Ingestion Layer:** Collects logs, metrics, and traces.
2. **Event Backbone:** Apache Kafka processes and streams telemetry events.
3. **Orchestration Layer:** Mastra coordinates multi-agent workflows.
4. **Incident Intelligence Layer:**Specialized AI agents perform analysis and reasoning.
5. **Institutional Memory Layer:** Qdrant stores semantic incident knowledge.
6. **Safety Layer:** Enkrypt AI validates recommendations.
7. **User Interface Layer:** SRE Dashboard enables monitoring and approvals.

8. Workflow

1. Logs, metrics, and traces are continuously ingested and streamed into Apache Kafka. The Mastra Incident Orchestrator consumes these events and triggers specialized agents.
2. The Anomaly Detection Agent identifies potential incidents and classifies severity levels.
3. The Historical Retrieval Agent queries Qdrant to retrieve similar incidents and organizational knowledge.

4. The Root Cause Analysis Agent correlates telemetry and historical information to determine probable causes.
5. The Remediation Agent proposes recovery steps and infrastructure runbooks.
6. Enkrypt AI validates every recommendation, detects hallucinations, and blocks unsafe actions.
7. Site Reliability Engineers review and approve high-risk recommendations through the dashboard.
8. After resolution, the Post-Mortem Agent automatically generates incident documentation and the Knowledge Update Agent stores new learnings back into Qdrant.
9. The platform continuously improves through this feedback loop.

9. Technology Stack

Mastra

Role:

- Multi-agent orchestration
- Workflow management
- State management
- Context sharing
- Branching logic
- Human-in-the-loop workflows

Qdrant

Role:

- Institutional memory
- Vector database
- Semantic search
- Retrieval-Augmented Generation
- Incident knowledge retrieval
- Continuous learning

Enkrypt AI

Role:

- Hallucination detection
- Safety evaluation
- Policy enforcement
- Reliability scoring
- Risk assessment
- Trustworthy recommendations

Additional Technologies

- Apache Kafka
- React.js
- Next.js
- Tailwind CSS
- Python
- FastAPI
- Kubernetes
- PostgreSQL
- OpenTelemetry
- Jaeger

10. Severity-Based Incident Workflow

1. Low Severity: Automated recommendations and logging.
2. Medium Severity: Deep analysis with Enkrypt AI validation.
3. High Severity: Safety validation and human approval.
4. Critical Severity: Automatic escalation, war room activation, notifications, and mandatory human approval.

11. Expected Impact

SentinelOps AI delivers significant operational benefits:

- Faster incident resolution
- Reduced downtime
- Improved service reliability
- Better knowledge retention
- Reduced manual documentation effort
- Safer remediation actions
- Lower operational costs
- Increased engineering productivity
- Continuous organizational learning
- Improved incident response maturity

12. Future Scope

- Predictive incident prevention
- Autonomous remediation execution
- Self-healing infrastructure
- Multi-cloud integrations
- Advanced forecasting models
- AI-driven capacity planning
- Intelligent runbook generation
- Enterprise collaboration integrations
- Cross-organization incident intelligence sharing

13. Conclusion

SentinelOps AI reimagines incident management as an intelligent, continuously learning system rather than a collection of disconnected monitoring tools.

By combining Mastra's multi-agent orchestration capabilities, Qdrant's institutional memory, and Enkrypt AI's safety validation framework, SentinelOps AI becomes a trustworthy engineering copilot capable of detecting incidents, retrieving organizational knowledge, recommending safe remediation actions, generating post-mortem reports, and continuously improving from every incident.

The platform enables engineering organizations to significantly reduce Mean Time to Resolution, preserve institutional knowledge, and build more reliable and resilient software systems through autonomous reasoning, safety guardrails, and continuous learning.